

Optimization

Why

This is a capital deployment optimization process aimed at optimizing the CAT Bond portfolio by maximizing expected return while minimizing portfolio tail risk (TVaR) and portfolio volatility (standard deviation), subject to portfolio allocation and risk constraints.

What

In CAT bond optimization, the optimizer generates multiple candidate portfolios. Each portfolio is evaluated using a Fitness Function, which determines how effective the portfolio is based on several factors such as:

- Expected Return
- TVaR
- Volatility (Standard Deviation)
- Portfolio Constraints

The Fitness Function acts as the scoring mechanism used by the optimizer to determine whether one portfolio performs better than another.

$$Fitness = w1(Return) - w2(TVaR) - w3(StdDev)$$

Risk Metrics & Inputs

The optimization process uses several risk and financial metrics, including:

- Premium
- AAL (Average Annual Loss)
- EL (Expected Loss)
- ROL (Rate on Line)
- Attachment Probability
- Exhaustion Probability
- TVaR
- Standard Deviation

Optimization Engine

The optimization engine uses Genetic Algorithm-based techniques, including:

- Portfolio Selection
- Mutation
- Crossover
- Fitness Evaluation
- Efficient Frontier Generation

Prerequisite for Optimization

1, Load Master Portfolio - The Master Portfolio contains the full universe of CAT bond deals available for investment. Deal name, Sponsor, Peril, Region, Attachment/Exhaustion, Limit, Spread, ROL, EL, AAL, Tvar, StdDev, Maturity Date, Inception Date, Expiry date

2, Load Existing Bound Portfolio - This has existing allocation, current deployed capital and list of deals, Company Share, Bound Limit, Bound Premium, PortfolioRiskMetrics, AUM

3, Perform Data Quality & Eligibility Checks

4, Define Portfolio Constraints. Each deal has Minimum deployable amount and Maximum deployable amount. Calculate Minshare and Max share

5, Define Optimization Objectives - Maximize Portfolio Return , Minimize Tvar and StdDev

6, These two Portfolios are combined to select Bound deals which are eligible for optimization

7, For each of the bound deals, there are Bound Premium, Bound Limit and Bound AAL, Min and Max AUM from where Min share and Max Share is calculated. Wherever the Min Share is Negative, those bonds need to be optimized. Max share is calculated using Premium (why)

8, We need to select the deals and mark them under Enable Exclude header, When Enable Exclude is True there is buy/ sell , when it is false (when MinShare is negative), then those deals picked up for optimization. When it is override which means Minshare=MaxShare then there is no optimization and we leave it as is

9, Generate multiple candidate portfolios and Evaluate Portfolio Fitness. $\text{Fitness} = w_1 (\text{Return}) - w_2 (\text{TVaR}) - w_3 (\text{StdDev})$; w_i - are optimization weights. Higher fitness = better portfolio.

10, Apply Genetic Algorithm Optimization - Select sample portfolios, perform crossover, apply mutation (Mutation Process randomly adjusts deal selection, portfolio weights, participation shares), Recalculate metrics, recalculate fitness, remove infeasible portfolios, Repeat until convergence

11, After Mutation Portfolio Constraints get validated and invalid portfolios get rejected.

12, Generate Efficient Frontier - Store optimized portfolios across different risk-return combinations. The Efficient Frontier represents the highest expected return for each risk level,

or minimum risk for each return level. X-axis $\rightarrow$ TVaR or StdDev, Y-axis $\rightarrow$ Expected Return

13, From the efficient frontier choose the preferred portfolio based on Return Target or capital efficiency

14, Generate final optimization outputs.